Margaret Wielatz

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EDUCATION

Master of Science in Mechanical Engineering **Expected May 2027**
Purdue University GPA: 3.80/4.0

- Coursework: Optimal Control and Estimation, Optimization, Machine Learning, Nonlinear Control, Probability

Bachelor of Science in Aerospace Engineering **May 2024**
University of Florida GPA: 3.82/4.0

- Honors: cum laude; Minor: Computer and Information Science and Engineering
- Coursework: Linear Control Systems, Stability and Control of Aircraft, Dynamics, Astrodynamics, Aerodynamics, Data Structures and Algorithms, Thermodynamics, Mechanics of Materials, Numerical Methods

WORK EXPERIENCE

Graduate Research Assistant **Aug. 2024 – present**
Purdue University *West Lafayette, IN*

- Validated adaptive control policies in a reconnaissance mission experiment with 85 participants. Compared policies tailored to the individual's inherent tendency to trust automation to a generalized control strategy.
- Developing an adaptive diagnostic agent using a decision tree to update thresholds for diagnosing HVAC faults.

Control Dynamics Intern **May 2023 – July 2023**
United Launch Alliance *Centennial, CO*

- Analyzed launch vehicle stability throughout the duration of flight by conducting a nominal frequency domain analysis using Nichols plots. Tuned gains to design a second stage autopilot with acceptable stability margins.
- Conducted a dispersion analysis using Monte Carlo simulations to analyze stability in varying flight conditions.
- Conducted spacecraft separation analysis to verify that vehicle orientation and attitude rate met requirements.

Stress Analysis Intern **Jan. 2022 – May 2022**
Gulfstream Aerospace *Savannah, GA*

- Extracted section views of aircraft components in CATIA and analyzed finite element analysis models of aircraft components using Patran for pre- and post-processing of Nastran static solutions.

LEADERSHIP AND PROJECTS

Autonomous Exploration TurtleBot Demonstration **Aug. 2025 – Dec. 2025**
Purdue University *West Lafayette, IN*

- Developed TurtleBot demo in Gazebo using Python/ROS 2. Used frontier-based exploration to map unknown areas by iteratively identifying frontier points and autonomously navigating to frontiers using a PID controller.
- Implemented A* for global planning. Used RRT* to plan local paths to avoid obstacles detected with LIDAR.

Chief Engineer – NASA Student Launch **Aug. 2022 – Apr. 2023**
Swamp Launch Rocket Team *Gainesville, FL*

- Led weekly technical meetings with 7 subteam leads and reviewed launch vehicle and payload designs, approved prototypes, and reviewed flight simulations. Placed 6th out of 51 teams in the Student Launch competition.
- Supervised payload and vehicle fabrication. Integrated the payload and avionics systems into the rocket.
- Developed technical reports that consisted of engineering drawings, design justifications, flight simulations, and testing results. Presented the design to a panel of NASA experts during 3 design reviews.

SKILLS

- **Programming/software:** MATLAB, Simulink, C++, Python, ROS 2, Linux, SolidWorks CSWA, Fusion360
- **Manufacturing:** milling machine, lathe, water jet, bandsaw, sheet metal tools, MIG welding